Enrollment No.: _____

Darshan UNIVERSITY

योगः कर्मसु कौशलम्

Darshan Institute of Computer Application

B.Sc. (IT) | Sem-3 | Winter-2023

Course Code	:	2106CS301	**Date**	:	04-11-2023
Course Name	:	Computer Network & Network Security	**Duration**	:	150 Minutes
			Total Marks	:	70

Instructions:

1. Attempt all the questions.
2. Figures to the right indicates maximum marks.
3. Make suitable assumptions wherever necessary.

Q.1	**(A)**	Explain application layer and list protocols working on it.	**4**
	(B)	Write full form of OSI? List seven layers of OSI reference model.	**3**
		OR	
		List all functions of network layer and explain any one in brief.	
	(C)	Draw and write OSI Reference Model in detail.	**7**
		OR	
		Write the functions of the Data-Link Layer.	
Q.2	**(A)**	List advantages and disadvantages of FDM.	**4**
	(B)	Define switching.	**3**
		OR	
		Explain TDM.	
	(C)	Explain circuit switching and packet switching.	**7**
		OR	
		Write a short note on multiplexing.	
Q.3	**(A)**	Draw IPv4 header format.	**4**
	(B)	Define the terms: i) IP address ii) host address iii) network address.	**3**
		OR	
		Classify IP addresses.	
	(C)	Demonstrate subnetting in detail.	**7**
		OR	
		A host has been assigned an IP address 192.168.17.9. Find the class, the number of addresses in the block, the first address, and the last address.	

Q.4	**(A)**	Explain unicast and multicast transmission.	**4**
	(B)	Clarify types of broadcast.	**3**
		OR	
		Describe casting and list its types.	
	(C)	Discuss routing types in detail.	**7**
		OR	
		Explain distance vector routing.	
Q.5	**(A)**	Differentiate authorization and authentication.	**4**
	(B)	List the useful Security Policies.	**3**
		OR	
		List the Security Methods.	
	(C)	Explain proxy server.	**7**
		OR	
		Write short note on VPN.	

***